

Client-Server Architecture

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■ Course structure Partial test

- 1st Partial = 30%
 - Autoevaluation = 5%
 - Theoretical evaluation = 40%
 - Practical evaluation (paper based?) = 55%

Agenda

- Client-Server Architecture
- Clients
- Servers
- Hand's on exploration
- Homework

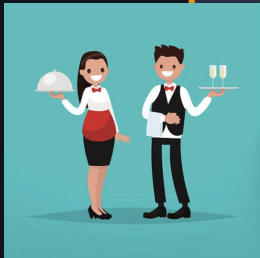
■ Client-Server Architecture

Client-Server Architecture

What is a server?
Let's begin with the
real world.

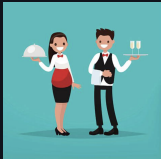
Those who work at a
restaurant or a bar
attending customers
and supplying them
with food and drink
requested.

as a role
(media)



Client-Server Architecture

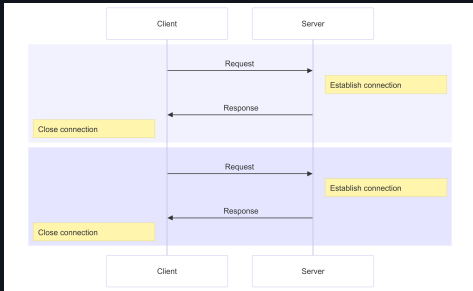
How do we interact with this kind of server?



1. A customer enter the establishment.
2. The customer tells the server what he/she wants to **request**.
3. The server goes on to fulfill the **request**.
4. Later the server returns with the customer **response**.

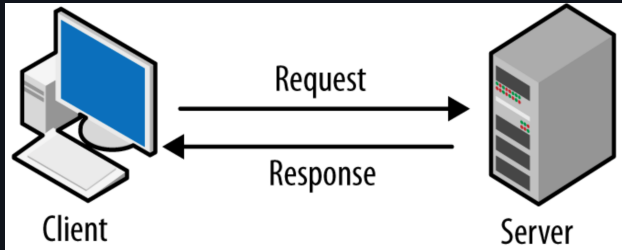
Client-Server Architecture

Does this diagram reflect this interaction between customers and servers?



Client-Server Architecture

We can stop thinking about people and replace them with devices:





Clients

A client is a program that, as part of its operation, relies on sending a request to a service made available by a server.

The easiest example I can think of is your web browser. But there are a lot other client examples:

- Videogame clients
- Email clients
- Streaming clients
- Weather clients

Servers

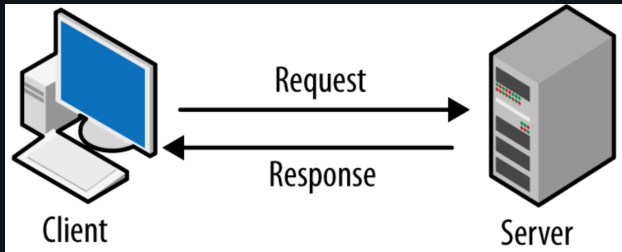
A server is a computer or software system that provides data, resources, or services to other computers called "clients" on a computer network.

Without servers, the clients don't have the data to display.



Servers

What are the benefits of a model like this?





Servers

Clients:

- Lightweight
- Can be replaced without impacting data access
- Just ask the information it needs

Servers:

- Store state (single source of truth)
- Can handle multiple clients at the same time
 - Responsible for authentication (who is the client) and authorization (what can the client access)

Servers

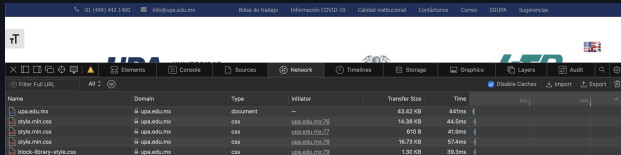
Servers tend to have this ominous image but that's just for reliability purposes.



Hand's on exploration

Open your browser (web client) and navigate to:
upa.edu.mx

Open your developer tools on network tab



The screenshot shows a web browser displaying the homepage of upa.edu.mx. The browser's address bar shows the URL. The developer tools are open, and the Network tab is selected. The Network tab displays a list of resources loaded by the page, including the HTML document and several CSS files. The table below represents the data shown in the Network tab.

Name	Domain	Type	Initiator	Transfer Size	Time	Size	Time
upa.edu.mx	upa.edu.mx	document	—	43.42 KB	441ms		
style.min.css	upa.edu.mx	css	upa.edu.mx:78	14.38 KB	44.5ms		
style.min.css	upa.edu.mx	css	upa.edu.mx:77	610 B	41.9ms		
style.min.css	upa.edu.mx	css	upa.edu.mx:78	16.73 KB	57.4ms		
block-library-style.css	upa.edu.mx	css	upa.edu.mx:78	1.90 KB	39.3ms		



Hand's on exploration

Each row is a request, we can analyze each:

The screenshot shows a web browser's developer tools network tab. On the left, a list of requests is shown, with the first one, 'upa.edu.mx', selected. The right pane displays the details for this request, including a summary and the full request headers.

Summary

- URL: <https://upa.edu.mx/>
- Status: 200
- Source: Network
- Address: 78.14.78.223:443

Request

```
method: GET
scheme: https
authority: upa.edu.mx
path: /
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Encoding: gzip, deflate, br, zstd
Accept-Language: en-US,en;q=0.9
Cache-Control: no-cache
Pragma: no-cache
Priority: u=0,1
Referer: https://upa.edu.mx/
Sec-Fetch-Dest: document
Sec-Fetch-Mode: navigate
Sec-Fetch-Site: none
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.116 (KHTML, like Gecko) Version/26.4 Safari/605.1.15
```



Hand's on exploration

Each row is a request, we can analyze each:

Name	Preview	Headers	Cookies	Sizes	Timing	Security	Response
upa.edu.mx	<pre>1 <!DOCTYPE html> 2 <html lang="es"> 3 <head> 4 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8"/> 5 <meta http-equiv="X-UA-Compatible" content="IE=edge"> 6 <meta name="viewport" content="width=device-width, initial-scale=1, maximum-scale=1"> 7 <link rel="profile" href="http://gmpg.org/xfn/11"/> 8 <link rel="pingback" href="https://upa.edu.mx/xmlrpc.php"/> 9 <meta name="robots" content="index, follow, max-image-preview:large, max-snippet:-1, max-video-preview:-1"/> 10 <!-- This site is optimized with the Yoast SEO plugin v7.2 - https://yoast.com/product/yoast-seo-wordpress/ --> 11 <title>Inicio - Universidad Politécnica de Aguascalientes</title> 12 <link rel="canonical" href="https://upa.edu.mx/" /> 13 <meta property="og:locale" content="es_MX"/> 14 <meta property="og:type" content="website"/> 15 <meta property="og:title" content="Inicio - Universidad Politécnica de Aguascalientes"/> 16 <meta property="og:description" content="Convocatoria Resultados Convocatoria Convocatoria iRegistrate en la Bolsa de Trabajo Darne de 17 alta Tramita tu Título del 23 de marzo al 01 de abril de 2026 Más Información SOLICITAR MAS INFORMACIÓN Maestría en Enseñanza de las Ciencias 18 iEstudia un posgrado en la UPA! Maestría en Ciencias en Innovación Tecnológica Convocatoria iRegistrate ya! CONOCE NUESTRO CATALOGO 19 [shellip]"/> 20 <meta property="og:url" content="https://upa.edu.mx/" /> 21 <meta property="og:site_name" content="Universidad Politécnica de Aguascalientes"/> 22 <meta property="article:publisher" content="https://www.facebook.com/UniversidadPolitecnicaAguascalientes"/> 23 <meta property="article:modified_time" content="2026-03-17T19:52:22+00:00"/> 24 <meta property="og:image" content="https://upa.edu.mx/wp-content/uploads/2021/02/Logo_UPA.jpg"/> 25 <meta property="og:image:width" content="1720"/> 26 <meta property="og:image:height" content="1048"/> 27 <meta property="og:image:type" content="image/jpeg"/> 28 <meta name="twitter:card" content="summary_large_image"/></pre>						

Hand's on exploration

But how does our browser knows to request all those resources?



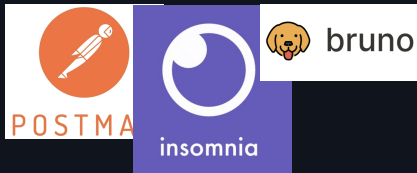
Hand's on exploration

But how does our browser knows to request all those resources?



Hand's on exploration

Let's use a different kind of web client:





Hand's on exploration

upa.edu.mx

GET upa.edu.mx

Params Body Headers Auth Vars Script Asset Tests Docs Settings

Query

Name	Value
Name	Value

Post

Response Headers Timeline Tests

200 OK 94ms 259.13KB

```
<!DOCTYPE html>
<html lang="es">
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=utf-8"/>

  <meta http-equiv="X-UA-Compatible" content="IE=edge">

  <meta name="viewport" content="width=device-width, initial-scale=1, maximum-scale=1">

  <link rel="profile" href="http://gmpg.org/xfn/11"/>

  <link rel="pingback" href="https://upa.edu.mx/xmlrpc.php"/>

  <meta name="robots" content="index, follow, max-image-preview:large, max-snippet:-1, max-video-preview:-1" />

  <!-- This site is optimized with the Yoast SEO plugin v27.2 - https://yoast.com/product/yoast-seo-wordpress/ -->
  <title>Inicio - Universidad Politécnica de Aguascalientes</title>
  <link rel="canonical" href="https://upa.edu.mx/" />
  <meta property="og:locale" content="es_MX" />
  <meta property="og:type" content="website" />
  <meta property="og:title" content="Inicio - Universidad Politécnica de Aguascalientes" />
  <meta property="og:description" content="Convocatoria Resultados Convocatoria Convocatoria ¡Regístrate en la Balsa de Trabajo! Dame de alta Tramita tu Título del 23 de marzo
```

Hand's on exploration

Let's try different content types:

- upa.edu.mx
- upa.edu.mx/wp-content/uploads/2024/03/banner-sitio-UPA_veda.png
- upa.edu.mx/wp-includes/css/dist/preferences/style.min.css?ver=6.9.4
- upa.edu.mx/wp-content/plugins/loftloader/assets/js/loftloader.min.js?ver=2025121501

Homework

Try and recreate more requests from different sites in postman